Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with DOE-guided process-development, physicochemical characterization, stability analysis, and scale-up documentation experience across polymer, thin-film, and electrochemical materials. Frames formulation/process questions through structure-property data, technical reports, and reproducible lab workflows.

CORE COMPETENCIES

Formulation & Process Development | Physicochemical Characterization | DOE/SPC | Thermal & Spectroscopic Analysis | Stability / Failure Modes | Scale-Up Documentation | GMP Documentation Awareness | Small-Molecule Materials

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process development for bio-based polymer systems, defining processing windows, material-quality attributes, and reproducible procedures relevant to scale-up.
- Use FT-IR, spectroscopy, and thermal analysis to connect processing conditions with physicochemical material properties and stability-relevant performance.
- Prepare technical reports and SOP-ready documentation to support process transfer, safety, and repeatable lab execution.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Synthesized and characterized 15+ functional organic/organometallic materials, evaluating structure-property relationships, solid-state/stability behavior, and degradation mechanisms using spectroscopic, electrochemical, and thermal methods.
- Executed stability and stress-test studies to identify oxidative degradation, interface delamination, and redox instability, then proposed material/process mitigations.
- Documented experimental methods, characterization results, and risk assessments in formats suitable for technical decision-making.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Developed CVD-based molecular coating processes and surface-preparation protocols, linking process parameters to morphology and device performance for scale-relevant thin-film workflows.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition under flow using DOE and in-situ QCM-D monitoring to improve coating uniformity and reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Polymer Process Development and Scale-Up Readiness FPD Adjacent

Led process-window definition for bio-based polymer systems, linking processing conditions to material quality attributes and reproducible procedures.

DOE, SPC, FT-IR, thermal analysis, SOPs

Functional Materials Stability Studies Physicochemical Assessment

Evaluated degradation and stability behavior of organic/organometallic materials under electrochemical and optical stress conditions.

UV-Vis-NIR, cyclic voltammetry, TGA, DSC, FT-IR

CVD Thin-Film Process Optimization Process Transfer

Developed controlled coating workflows and surface-preparation protocols, translating process parameters into material morphology and performance.

CVD, QCM-D, spectroscopy, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder
2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)
2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process Development: DOE, SPC, process windows, scale-up documentation, SOPs, technical reports, safety protocols
Characterization: FT-IR, UV-Vis-NIR, TGA, DSC, cyclic voltammetry, spectroelectrochemistry, NMR, GPC, LC/GC-MS exposure
Materials & Formulation Adjacent: Polymer systems, small-molecule organic materials, thin films, coatings, solid-state/stability assessment, physicochemical properties
Regulatory-Adjacent: GMP documentation awareness, method reproducibility, risk assessments, development reports, cross-functional handoffs

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)